NO. OF FUNCTIONAL FEATURES INCLUDED IN THE SOLUTION

Project Title

Rising Waters

Project Subtitle

AI-Powered Flood Prediction System using Machine Learning

Introduction

Rising Waters is a Machine Learning-based web application developed to predict flood risk using rainfall and environmental parameters. The application provides accurate flood prediction through an interactive Streamlit dashboard.

Functional Features

1. User-Friendly Dashboard

The application provides a simple and professional web interface where users can easily enter environmental parameters.

2. Rainfall Data Input

Users can enter annual rainfall and seasonal rainfall values for prediction.

3. Weather Parameter Input

The system accepts temperature, humidity, and cloud cover values for analysis.

4. Input Validation

The application validates user input before processing to ensure accurate predictions.

5. Machine Learning Prediction

The trained XGBoost Machine Learning model predicts flood risk based on the provided environmental data.

6. Prediction Result Display

The application displays whether the flood risk is HIGH or LOW in a clear and understandable format.

7. Fast Processing

The prediction is generated within a few seconds, providing a quick response to the user.

8. Accurate Prediction

The trained model provides reliable flood prediction using historical rainfall and weather data.

9. Streamlit Web Application

The entire system is deployed as an interactive web application using Streamlit.

10. GitHub Integration

The project source code, documentation, and deployment files are maintained in a GitHub repository for version control.

Benefits

• Easy-to-use interface
• Accurate flood prediction
• Fast response time
• Better disaster preparedness
• Supports government authorities
• Scalable and maintainable system

Conclusion

The Rising Waters application includes all essential functional features required for an AI-powered Flood Prediction System. These features make the application reliable, user-friendly, and suitable for real-world disaster management.